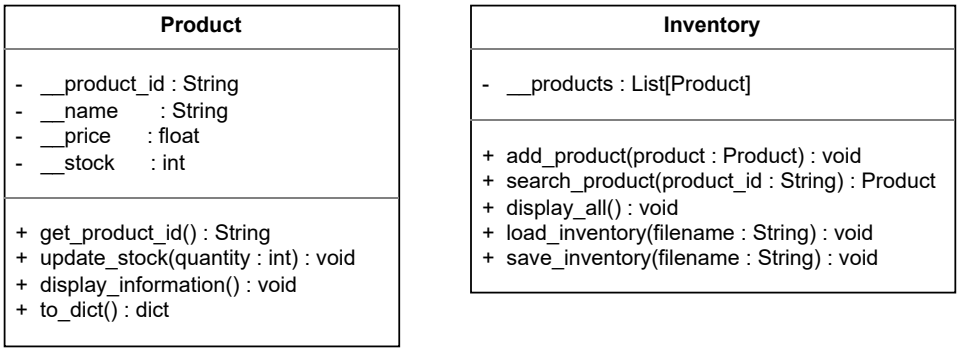
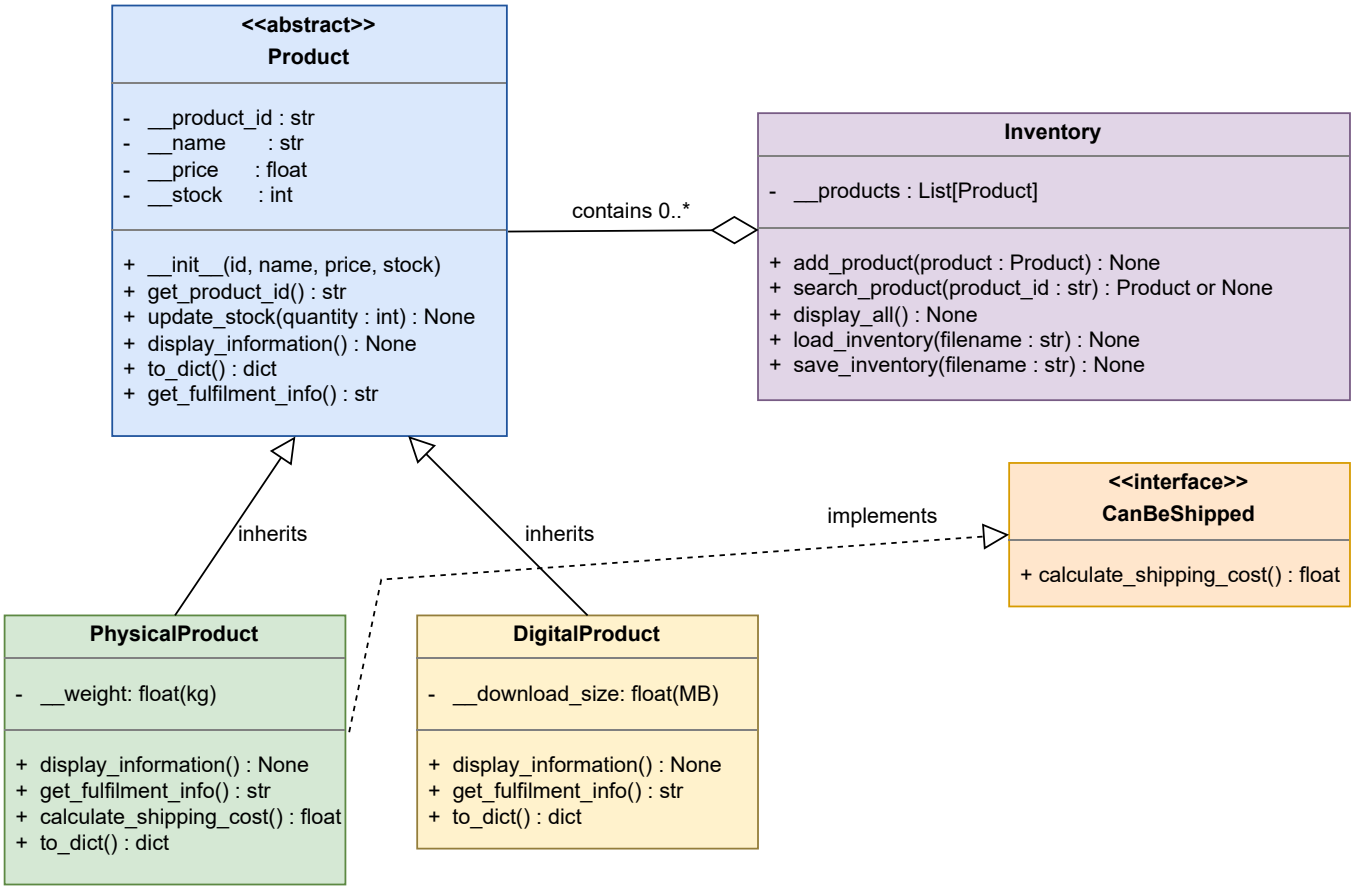
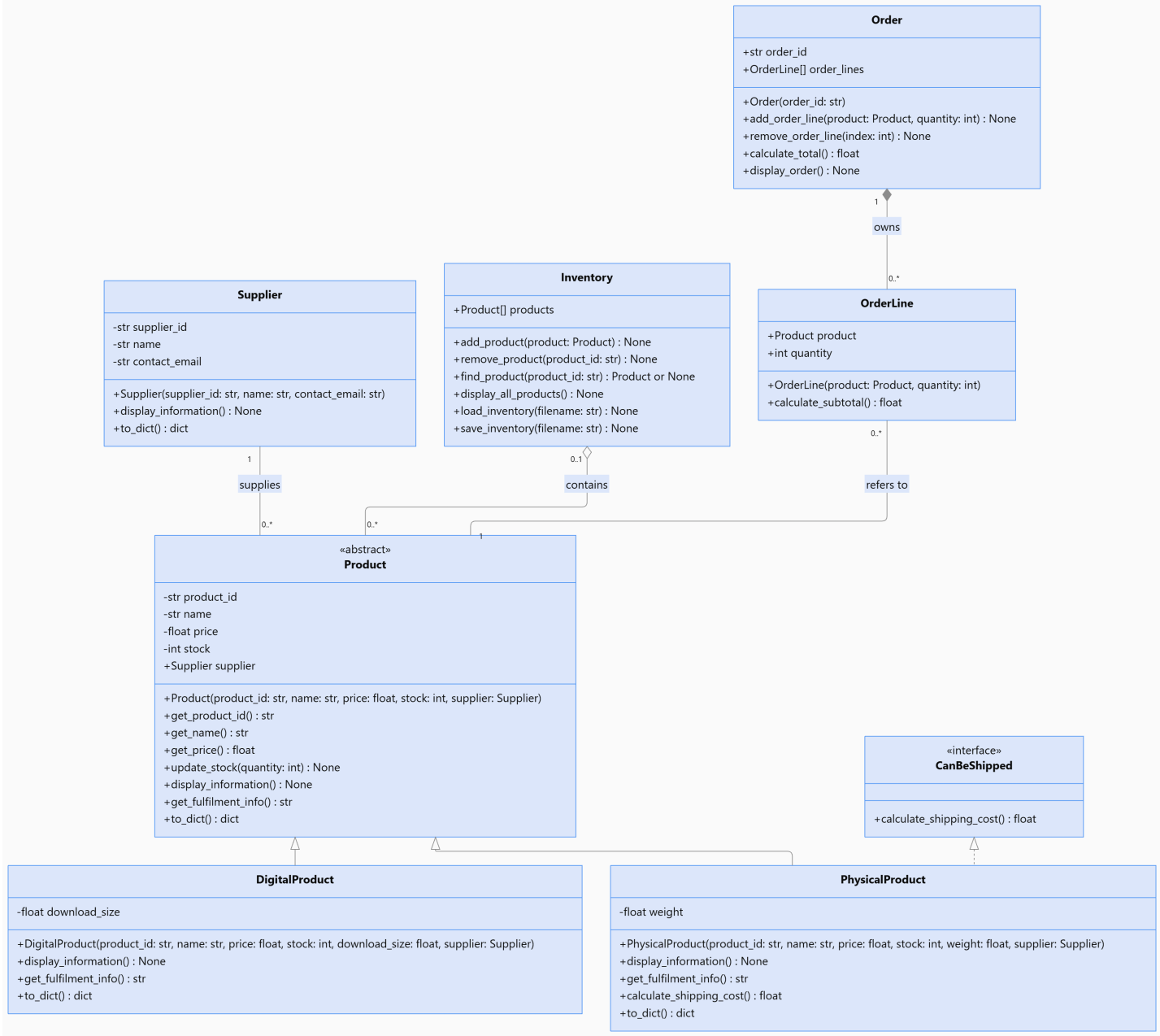
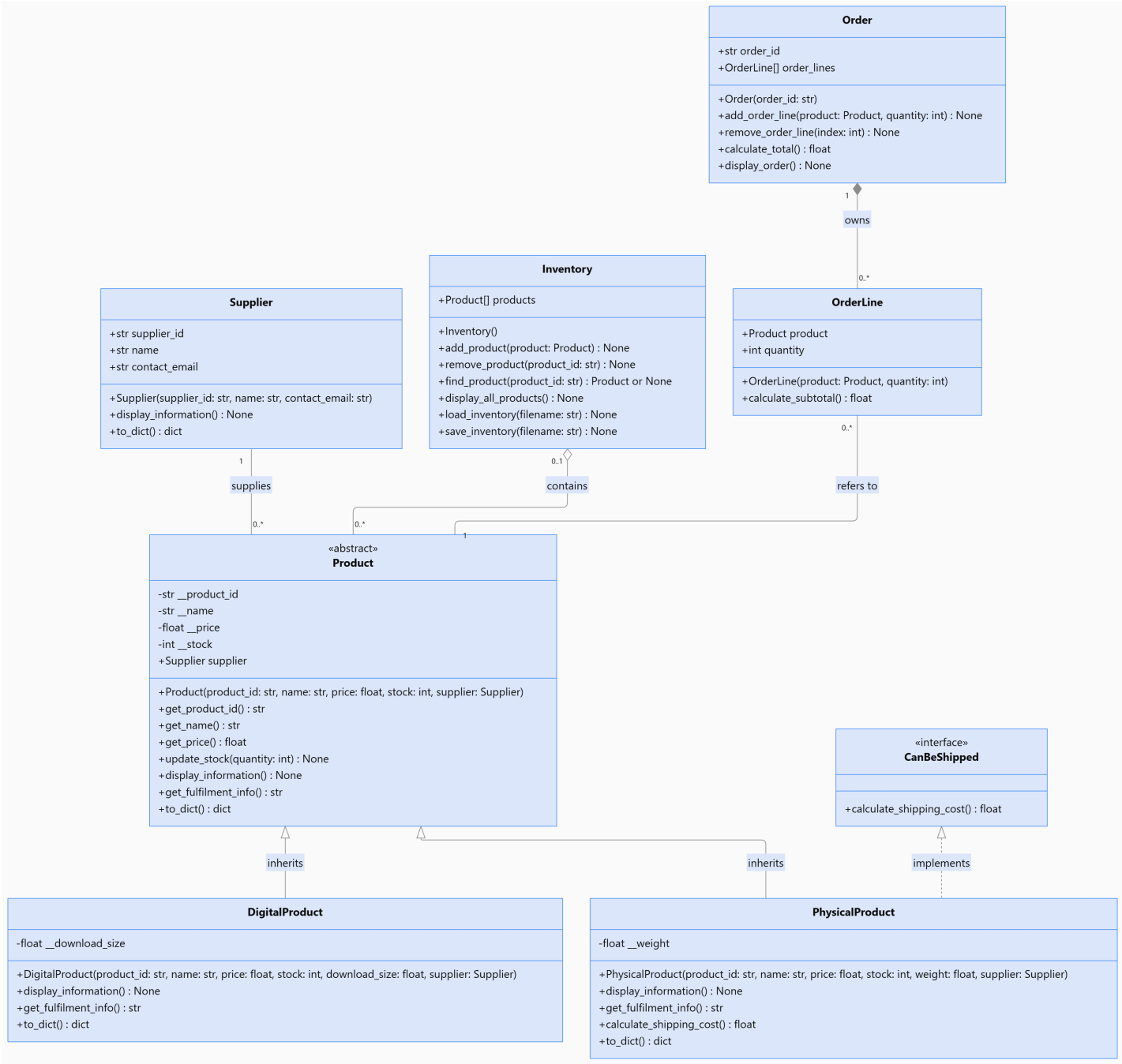

Product
- __product_id : String
- __name : String
- __price : float
- __stock : int
+ get_product_id() : String
+ update_stock(quantity : int) : void
+ display_information() : void
+ to_dict() : dict
Inventory
- __products : List[Product]
+ add_product(product : Product) : void
+ search_product(product_id : String) : Product
+ display_all() : void
+ load_inventory(filename : String) : void
+ save_inventory(filename : String) : void


<<abstract>>
Product
- __product_id : str
- __name : str
- __price : float
- __stock : int
+ __init__(id, name, price, stock)
+ get_product_id() : str
+ update_stock(quantity : int) : None
+ display_information() : None
+ to_dict() : dict
+ get_fulfilment_info() : str
contains 0..*
Inventory
- __products : List[Product]
+ add_product(product : Product) : None
+ search_product(product_id : str) : Product or None
+ display_all() : None
+ load_inventory(filename : str) : None
+ save_inventory(filename : str) : None
inherits
inherits
implements
<<interface>>
CanBeShipped
+ calculate_shipping_cost() : float
PhysicalProduct
- __weight: float(kg)
+ display_information() : None
+ get_fulfilment_info() : str
+ calculate_shipping_cost() : float
+ to_dict() : dict
DigitalProduct
- __download_size: float(MB)
+ display_information() : None
+ get_fulfilment_info() : str
+ to_dict() : dict


Order
+str order_id
+OrderLine[] order_lines
+Order(order_id: str)
+add_order_line(product: Product, quantity: int) : None
+remove_order_line(index: int) : None
+calculate_total() : float
+display_order() : None
1
owns
0..*
Supplier
-str supplier_id
-str name
-str contact_email
+Supplier(supplier_id: str, name: str, contact_email: str)
+display_information() : None
+to_dict() : dict
Inventory
+Product[] products
+add_product(product: Product) : None
+remove_product(product_id: str) : None
+find_product(product_id: str) : Product or None
+display_all_products() : None
+load_inventory(filename: str) : None
+save_inventory(filename: str) : None
OrderLine
+Product product
+int quantity
+OrderLine(product: Product, quantity: int)
+calculate_subtotal() : float
0..*
1
supplies
0..*
0..1
contains
0..*
1
refers to
«abstract»
Product
-str product_id
-str name
-float price
-int stock
+Supplier supplier
+Product(product_id: str, name: str, price: float, stock: int, supplier: Supplier)
+get_product_id() : str
+get_name() : str
+get_price() : float
+update_stock(quantity: int) : None
+display_information() : None
+get_fulfilment_info() : str
+to_dict() : dict
«interface»
CanBeShipped
+calculate_shipping_cost() : float
DigitalProduct
-float download_size
+DigitalProduct(product_id: str, name: str, price: float, stock: int, download_size: float, supplier: Supplier)
+display_information() : None
+get_fulfilment_info() : str
+to_dict() : dict
PhysicalProduct
-float weight
+PhysicalProduct(product_id: str, name: str, price: float, stock: int, weight: float, supplier: Supplier)
+display_information() : None
+get_fulfilment_info() : str
+calculate_shipping_cost() : float
+to_dict() : dict


Order
+str order_id
+OrderLine[] order_lines
+Order(order_id: str)
+add_order_line(product: Product, quantity: int) : None
+remove_order_line(index: int) : None
+calculate_total() : float
+display_order() : None
1
owns
0..*
Inventory
+Product[] products
+Inventory()
+add_product(product: Product) : None
+remove_product(product_id: str) : None
+find_product(product_id: str) : Product or None
+display_all_products() : None
+load_inventory(filename: str) : None
+save_inventory(filename: str) : None
Supplier
+str supplier_id
+str name
+str contact_email
+Supplier(supplier_id: str, name: str, contact_email: str)
+display_information() : None
+to_dict() : dict
OrderLine
+Product product
+int quantity
+OrderLine(product: Product, quantity: int)
+calculate_subtotal() : float
0..*
1
supplies
0..*
0..1
contains
0..*
refers to
1
«abstract»
Product
-str __product_id
-str __name
-float __price
-int __stock
+Supplier supplier
+Product(product_id: str, name: str, price: float, stock: int, supplier: Supplier)
+get_product_id() : str
+get_name() : str
+get_price() : float
+update_stock(quantity: int) : None
+display_information() : None
+get_fulfilment_info() : str
+to_dict() : dict
«interface»
CanBeShipped
+calculate_shipping_cost() : float
inherits
inherits
implements
DigitalProduct
-float __download_size
+DigitalProduct(product_id: str, name: str, price: float, stock: int, download_size: float, supplier: Supplier)
+display_information() : None
+get_fulfilment_info() : str
+to_dict() : dict
PhysicalProduct
-float __weight
+PhysicalProduct(product_id: str, name: str, price: float, stock: int, weight: float, supplier: Supplier)
+display_information() : None
+get_fulfilment_info() : str
+calculate_shipping_cost() : float
+to_dict() : dict